have time to line up, and they will be stuck in any old arrangement, with no order to the arrangement of the atoms. We call materials like this as amorphous. At this stage, glass is linear in arrangement inorganic in nature and has a structure very similar to glass and they are considered as polymers.

In a commercial glass plant, sand is mixed with waste glass (obtained from recycling collections), soda ash (sodium carbonate) and limestone (calcium carbonate) and heated in a furnace. The soda ash reduces the sand's melting point and produces a kind of glass that would dissolve in water. The limestone is added to stop that happening. The end product is called soda-lime-silica glass. It is the ordinary glass we mostly use in our daily life.

Usually, other chemicals are added to change the appearance or properties of the finished glass. For example, iron and chromium based chemicals are added to the molten sand to make green-tinted glass.

Oven-proof borosilicate glass (widely sold under the trademark PYREX) is made by adding boron oxide to the molten mixture.

Adding lead oxide makes from a sandwich or laminate of multiple layers of glass and plastic bonded together.

Bulletproof glass is made a formed by compressing different layers of glass and plastic one after another.

Toughened glass used in car winds hields is made by cooling molten glass very quickly to make it much harder.

When silver iodide is added to molten glass the glass obtained becomes dark glasses when exposed to sunlight or any other light. These glasses are used in the production of lenses for eyes and spectacles

Points to Remember

- Fibres are long strands of molecules interwoven to form linear, string-like structures that can be woven, knitted, matted or bonded.
- Fibres can be natural or synthetic.
- Our increasing consumption and production of waste has negative impacts on humans, animals and the environment.
- Nylon is a synthetic fibre made from petrochemicals oil and gas.
- Polycot is a mixture of polyester and cotton, Polywool is a mixture of polyester and wool.
- Synthetic fibres melt on burning while cotton fibres catch fire on burning.
- Polymers are very long chains made of repeating smaller molecules called monomers that are joined together with different types of bonds.
- Plastics that get deformed easily on heating and can be bent are known as thermoplastics.
- Plastics that do not get deformed easily on heating and cannot be bent are known as thermosetting plastics.
- There are many different types of plastics. Some of which are safe or unsafe for us. We can find out what type of plastic an item is by identifying the resin code or acronym found on it.
- Poly Lactic Acid or polylactide is a biodegradable and bioactive thermoplastic.
- The management of plastics can be considered by the 5R Principle – **Refuse, Reduce, Reuse, Recycle** and `**Recover** as well as to **Landfill**.
- Ideonellasakaiensis 201-F6 is a bacteria discovered by scientists that digests plastic resin code #1 PET used to make single-use drinks bottles.

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